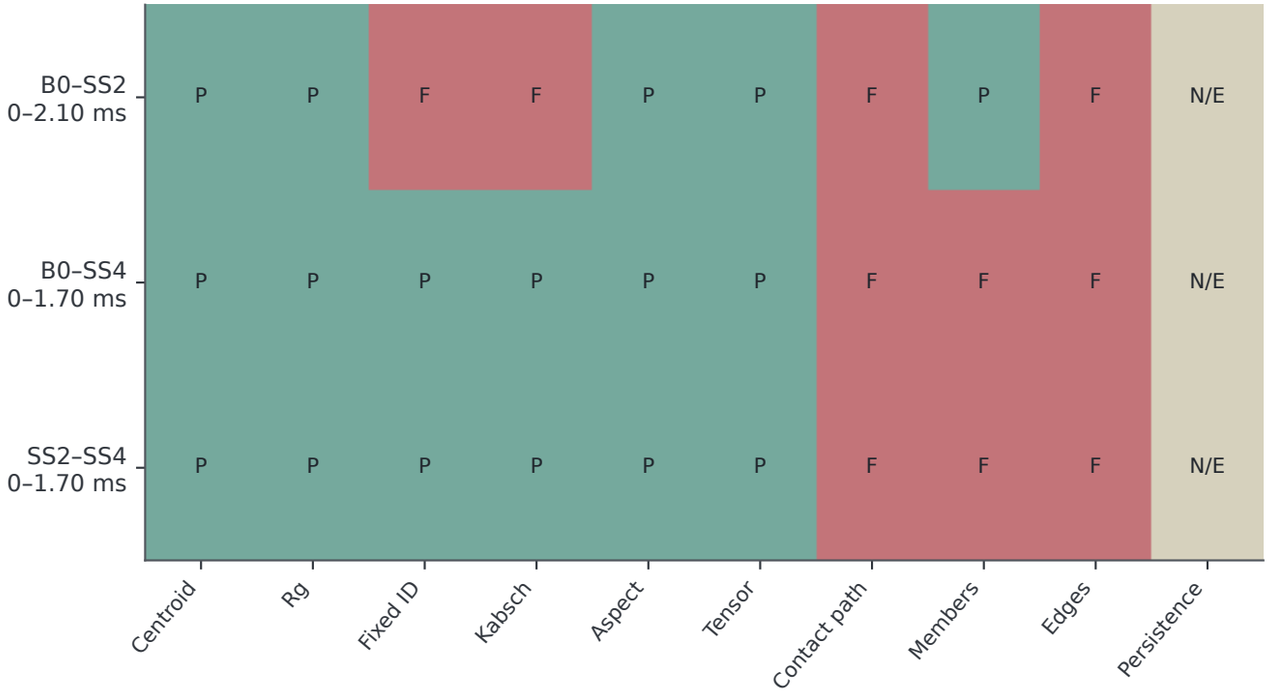
F10F | Small Steps is not admitted as a production numerical mode

a Contact outcome by original case

Case	Mode	Rejected (ms)	Residual (nm)	17.8 nm gate
O / 1102	B0	4.9	17.831	FAIL
O / 1102	SS2	0.005	20.920	FAIL
O / 1102	SS4	0.0025	20.361	FAIL
Prep. / 1104	B0	2.24	17.831	FAIL
Prep. / 1104	SS2	2.11	18.481	FAIL
Prep. / 1104	SS4	1.745	18.281	FAIL

b Strict common-prefix morphology gates



c Observed failure mechanism

Same O / 1102 starting coordinates:
Prediction overlap: B0 85.544 nm → SS2 50.535 nm
→ SS4 32.978 nm

Smaller step reduces the predicted penetration.
Fewer Jacobi passes leave 20.920 / 20.361 nm residual.
The original 17.8 nm limit rejects both first substeps.

On the same B0 rejected proposal, one round leaves
66.191 nm (O) / 55.276 nm (preparation);
12 rounds still leave 17.831 nm.
This local test is not a new physical trajectory.

d Method admission and scope

Engineering checks: PASS
Original historical defaults and results: preserved
Small Steps contact windows: FAIL (4 / 4)
Group centroid agreement: small on accepted prep prefixes
Full morphology admission: NOT PASSED
Production speed / small batch: NOT ADMITTED

Supported evidence: two fixed original cases, 32 bubbles.
Not supported: dense long-time production, independent
generalization, morphology training truth or calibrated shells.

Next: local lubrication convergence and near-field audit;
assess contact prediction only with unchanged acceptance.
EXPLORATORY / PHYSICS_UNCALIBRATED / training_ready=false

P = passed only the displayed common prefix; F = failed; N/E = right-censored. O / 1102 SS morphology is not evaluated: no accepted nonzero substep.